

Multi-Agent Scientific Paper Review

Powered by Phi-3 Mini · May 03, 2026 19:20

File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	qwen3:8b

■ **AI-ASSISTED REVIEW NOTICE:** This report was generated by an artificial intelligence system using Phi-3 Mini as the language model. It is intended as a supplementary tool to assist human peer reviewers. All findings must be validated by a qualified domain expert before editorial decisions are made. The AI may miss context-specific nuances and should not be used as the sole basis for acceptance or rejection.

Editorial Decision

MAJOR REVISION

Weighted Score: 3.3 / 5.0
Confidence: 60%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	4.0	Good	■■■■■
Manuscript Structure	4.2	Good	■■■■■
Methodology & Equations	2.5	Weak	■■■■■
Statistical Analysis	3.5	Good	■■■■■
Results & Evidence	—	Skipped	—
Discussion & Conclusions	—	Skipped	—
Figures & Tables	4.0	Good	■■■■■
Scientific Writing	3.0	Acceptable	■■■■■
References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	3.5	Good	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 4.0 / 5.0 | Confidence: 65%

Strengths:

- [TITLE] Title detected and available for review.
 - [ABSTRACT] Abstract detected and available for review.
 - [KEYWORDS] Keywords detected and available for review.
-

StructureReviewer

Manuscript structure and logical coherence

Score: 4.2 / 5.0 | Confidence: 65%

Strengths:

- The manuscript follows a detectable progression: introduction -> methodology -> experiments -> discussion -> conclusions.
 - The introduction appears to establish the problem/objective before the technical sections.
 - The technical middle sections contain method/framework/model signals, supporting a methodological reading.
 - Validation/results signals are present after the methodological material.
 - The discussion appears to refer back to results, which supports structural coherence.
 - A substantive conclusion section was detected.
-

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 2.5 / 5.0 | Confidence: 55%

Weaknesses:

- Could not parse model response — raw output stored

Recommendations:

- Re-run this agent or check model output
-

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: 3.5 / 5.0 | Confidence: 60%

Strengths:

- The manuscript reports multiple statistical validation elements, including independent runs, nonparametric tests, correction procedures, confidence intervals, and performance/dispersion metrics.

Weaknesses:

- Could not parse model response — raw output stored

Recommendations:

- Re-run this agent or check model output
-

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: 4.0 / 5.0 | Confidence: 62%

Strengths:

- 6 figure references were detected in the manuscript.
- 10 table references were detected in the manuscript.
- 52 sequentially numbered equations were detected, suggesting a substantial mathematical formulation.

Weaknesses:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Minor Comments:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Recommendations:

- Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
-

ResultsReviewer

Results validity and empirical support

Score: **0.0 / 5.0** | Confidence: 0%

Weaknesses:

- Could not parse model response — raw output stored

Recommendations:

- Re-run this agent or check model output
-

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **0.0 / 5.0** | Confidence: 0%

Weaknesses:

- Could not parse model response — raw output stored

Recommendations:

- Re-run this agent or check model output
-

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.0 / 5.0** | Confidence: 55%

Strengths:

- The manuscript contains enough extracted text for a representative writing review.
- Technical terminology and quantitative reporting signals are present in the extracted text.

Weaknesses:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Minor Comments:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Recommendations:

- Review representative paragraphs manually for transition quality, undefined acronyms, and overly long sentences.
 - Ensure technical terms are defined on first use and used consistently across sections.
-

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 55%

Weaknesses:

- Could not parse model response — raw output stored

Minor Comments:

- The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Recommendations:

- Re-run this agent or check model output
-

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **3.5 / 5.0** | Confidence: 60%

Strengths:

- No clear human/animal-subject signal was detected in the parsed manuscript.

Weaknesses:

- Conflict-of-interest declaration was not detected.

Major Comments:

- Conflict-of-interest declaration was not detected.

Recommendations:

- Include a conflict-of-interest declaration.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

[1] Conflict-of-interest declaration was not detected.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

[1] The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

[2] The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

[3] The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Specific Improvement Recommendations

1. Re-run this agent or check model output
2. Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
3. Review representative paragraphs manually for transition quality, undefined acronyms, and overly long sentences.
4. Ensure technical terms are defined on first use and used consistently across sections.
5. Include a conflict-of-interest declaration.

This review was generated automatically by the Multi-Agent Paper Review System using Phi-3 Mini (Microsoft) running locally via Ollama. No manuscript content was transmitted to external servers. Rubrics based on Elsevier, IEEE, and Emerald Q1 peer review standards. This document does not constitute an official editorial decision.